

Data Science

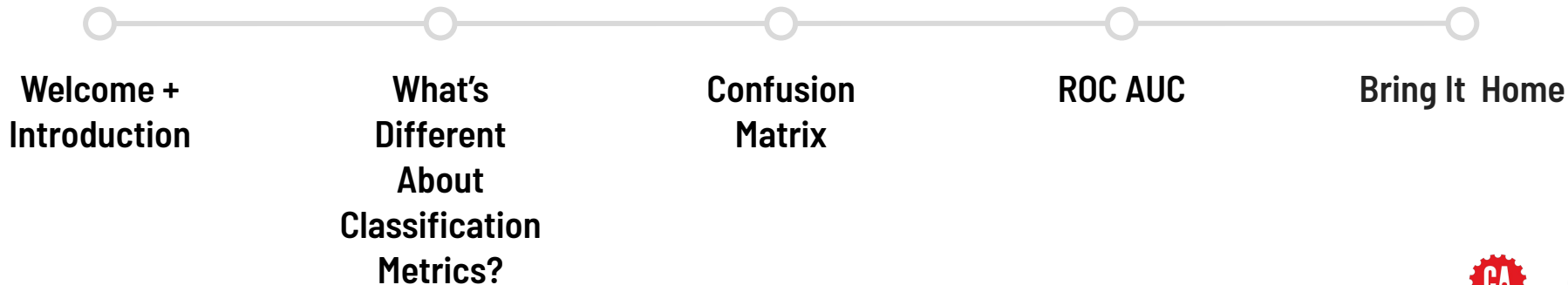
Classification Metrics

Machine Learning Fundamentals



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Compare regression and classification metrics.

2

Define common metrics and the confusion matrix.

3

Describe the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) score.

What's Different About Classification Metrics?

Classification vs Regression Metrics

Why do we need a whole new set of metrics?

For **regression**, we're primarily concerned with how close our predictions are to the truth. This was generally some clever summarization of the **residuals**. That is, some measurement of $y - \hat{y}$.

But for **classification**, we don't have residuals! There's no numeric difference between category variables.

Classification Metrics: Framing

Up until now, we've only used **accuracy** as our evaluation metric. But in many cases, especially those with **class imbalance**, it can be a poor (or misleading!) metric.

For the first time, we need to consider the **actual context** of the problem when deciding on the most important evaluation metric.



Some Quick Vocab

Suppose you build a logistic regression model to predict whether or not someone will vote in the upcoming election. You make predictions for 100 people and then follow up with them:

- ☐ There are 40 people you predicted to vote who actually voted.
- ☐ There are 20 people you predicted to vote but did not vote.
- ☐ There are 15 people you predicted to stay home who actually voted.
- ☐ There are 25 people you predicted to stay home who did not vote.

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*These are **true positives***
- ☐ There are 20 people you predicted to vote but did not vote.
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- ☐ There are 25 people you predicted to stay home who did not vote.
*These are **true negatives***



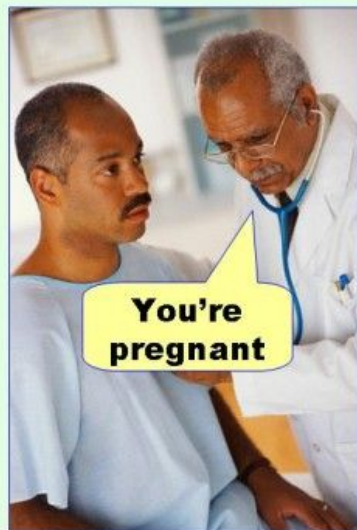
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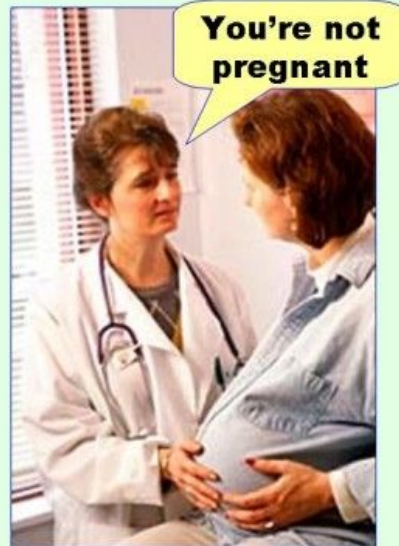
- ☐ There are 40 people you predicted to vote who actually voted.
*These are **true positives***
- ☐ There are 20 people you predicted to vote but did not vote.
*These are **false positives***
- ☐ There are 15 people you predicted to stay home who actually voted.
*These are **false negatives***
- ☐ There are 25 people you predicted to stay home who did not vote.
*These are **true negatives***

Another Name for These

Type I error
(false positive)



Type II error
(false negative)



A Simple Mnemonic

F False
positive

One line =>

Type I

NI False
negative

Two lines =>

Type II

Aside on Type I and II Errors

In the fields of inference and decision theory, Type I and Type II errors are used more commonly. We can actually match some terms up here:

	H_0 True	H_0 False
Fail to Reject H_0		
Rejected H_0		

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Fail to Reject H_0	Good!	Type II Error
Rejected H_0	Type I Error	Good!

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Rejected H_0	Type I Error	Good!

	H_0 True	H_0 False
Fail to Reject H_0	(No catchy name)	β
Rejected H_0	α	Power (K)

Different Error Types: Cancer Detection

You are a physician doing cancer screens. You conduct a screen and must either report “cancer” (a positive) or “no cancer” (a negative).

- ☐ What would be considered a false positive?
- ☐ What would be considered a false negative?
- ☐ Which is worse in this case?

Different Error Types: Advertising

You run a digital ad agency and you're deploying a model that determines whether or not you should show an ad to the visitor of a website based on metadata you've collected about this user. **You either show the specific ad for your client (positive) or you show a random other ad from your portfolio (negative).** Your goal is that the user will click this ad.

- ☐ What would be considered a false positive?
- ☐ What would be considered a false negative?
- ☐ Which is worse in this case?

Confusion Matrix

The Confusion Matrix

A **confusion matrix** is a two-way frequency table showing the counts of all of your outcomes in terms of predicted vs actual:

	Actual Positive	Actual Negative
Predicted Positive		
Predicted Negative		

(Note that some software, including Scikit-Learn, will flip the rows and columns. Pay attention!)

The Confusion Matrix

A **confusion matrix** is a two-way frequency table showing the counts of all of your outcomes in terms of predicted vs actual:

	Actual Positive	Actual Negative
Predicted Positive	True Positive	False Positive
Predicted Negative	False Negative	True Negative

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The Confusion Matrix

A **confusion matrix** is a two-way frequency table showing the counts of all of your outcomes in terms of predicted vs actual:

	Actually Voted	Did Not Vote
Predicted to Vote	40	20
Predicted Stay Home	15	25

(Note that some software, including Scikit-Learn, will flip the rows and columns. Pay attention!)

Classification Metrics

The confusion matrix gives us a bird's-eye view of performance, but we can also use it to calculate some single-number summaries:

- ☐ Accuracy
- ☐ Misclassification rate
- ☐ Sensitivity
- ☐ Specificity
- ☐ Precision
- ☐ Recall



Accuracy

The **accuracy** of a model is the proportion of correct predictions it makes.

The **misclassification rate** is the opposite - the proportion it gets incorrect!

You can a baseline the accuracy for your model by predicting the most common category for every data point. So if the most common class makes up 70% of your data, then your model needs to be at least 70% accurate.

Accuracy

The **accuracy** of a model is the proportion of correct predictions it makes.

The **misclassification rate** is the opposite - the proportion it gets incorrect!

	Actually Voted	Did Not Vote
Predicted to Vote	40	20
Predicted Stay Home	15	25

What is the accuracy of our voter model?

Accuracy

	Actually Voted	Did Not Vote
Predicted to Vote	40	20
Predicted Stay Home	15	25

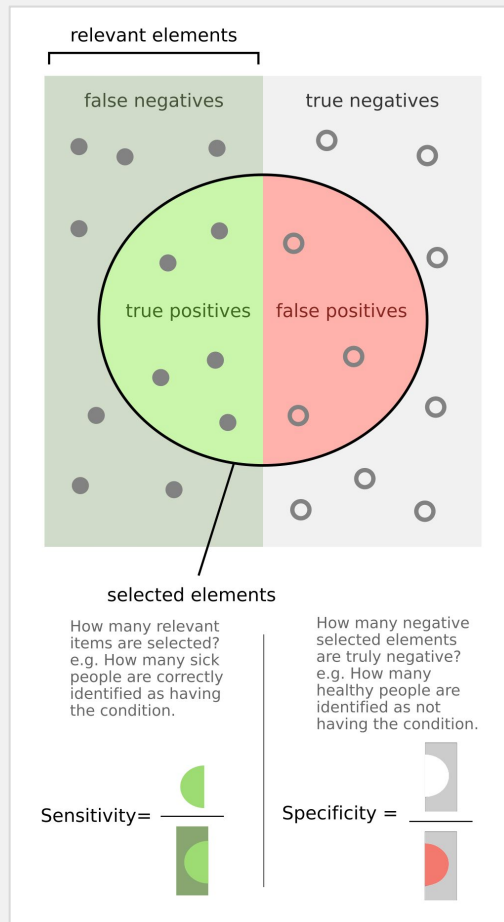
$$\text{Acc} = \frac{40+25}{40+20+15+25} = \frac{65}{100} = 0.65$$

$$\text{Miscl. Rate} = \frac{20+15}{40+20+15+25} = \frac{35}{100} = 1 - 0.65 = 0.35$$

Sensitivity & Specificity

The **sensitivity** is the **true positive rate** - that is, among all actual positives, what proportion did we detect?

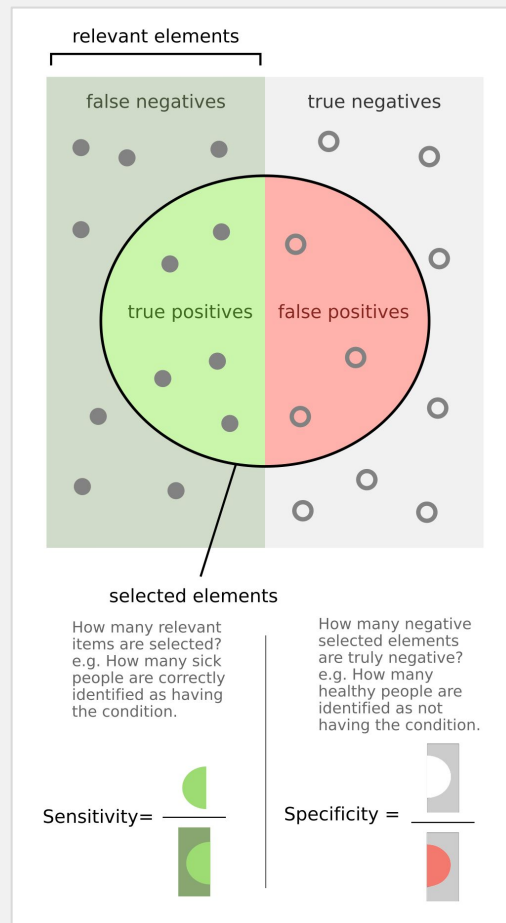
The **specificity** is the **true negative rate** - that is, among all true negatives, what proportion did we detect?



Sensitivity & Specificity

What are the sensitivity and specificity of our model?

	Actually voted	Did not vote
Pred to vote	40	20
Pred stay home	15	25



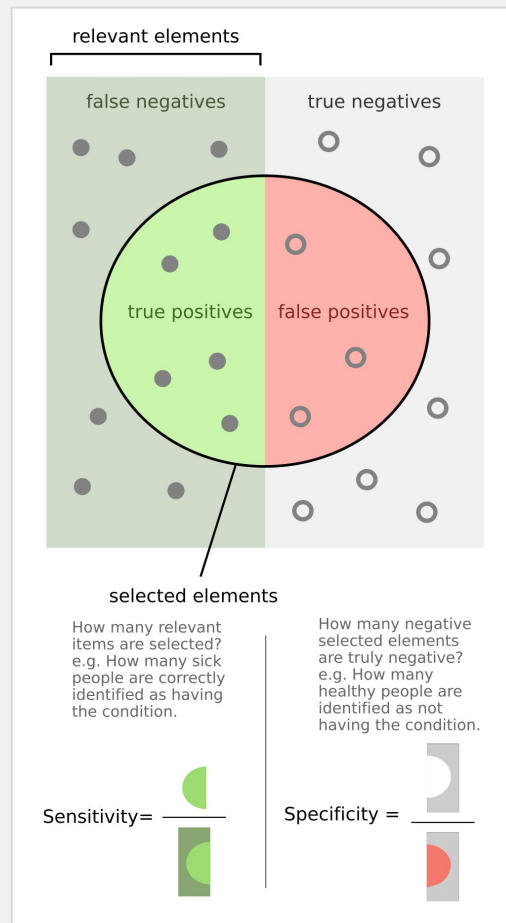
Sensitivity & Specificity

What are the sensitivity and specificity of our model?

	Actually voted	Did not vote
Pred to vote	40	20
Pred stay home	15	25

$$\text{Sensitivity} = \frac{40}{40+15} = \frac{40}{55} \approx 0.727$$

$$\text{Specificity} = \frac{25}{20+25} = \frac{25}{45} \approx 0.556$$



Sensitivity vs Specificity

An airport security scanner classifies passengers as 'dangerous' or 'safe'.

The confusion matrix, showing the scanner's performance on a testing set of 250 passengers, is on the right.

Calculate the sensitivity and specificity of the scanner, and explain these figures using words that could be easily understood by a passenger.

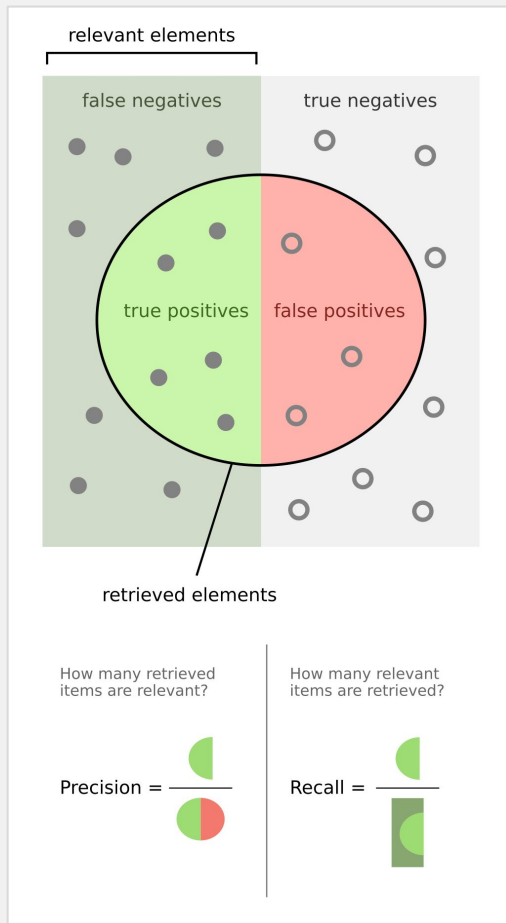
What's your assessment of how well the scanner does its job?

		Predicted Value	
		Dangerous	Safe
Actual Value	Dangerous	3	0
	Safe	200	47

Precision & Recall

The **precision**, aka **positive predictive value** - is, among all predicted positives, what proportion were correct?

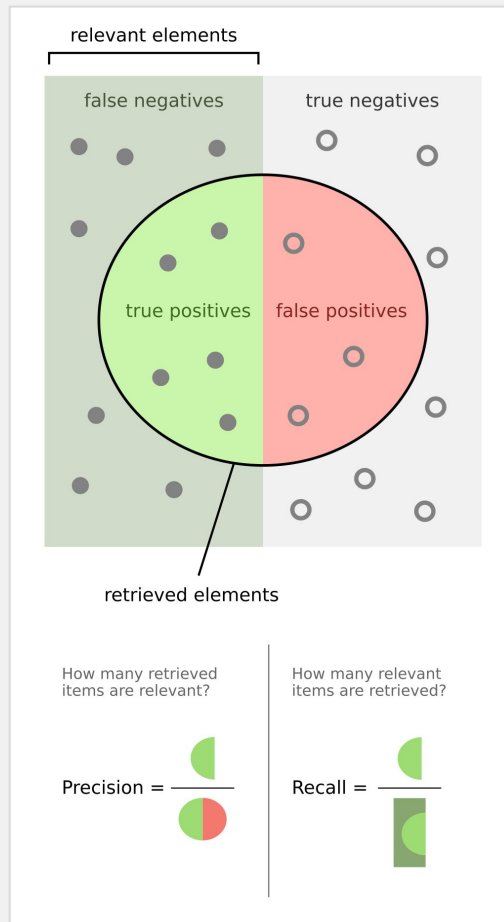
The **recall** is the **true positive rate** - that is... the same as **sensitivity**.



Precision & Recall

What are the precision and recall of our model?

	Actually Voted	Did Not Vote
Pred to Vote	40	20
Pred Stay Home	15	25



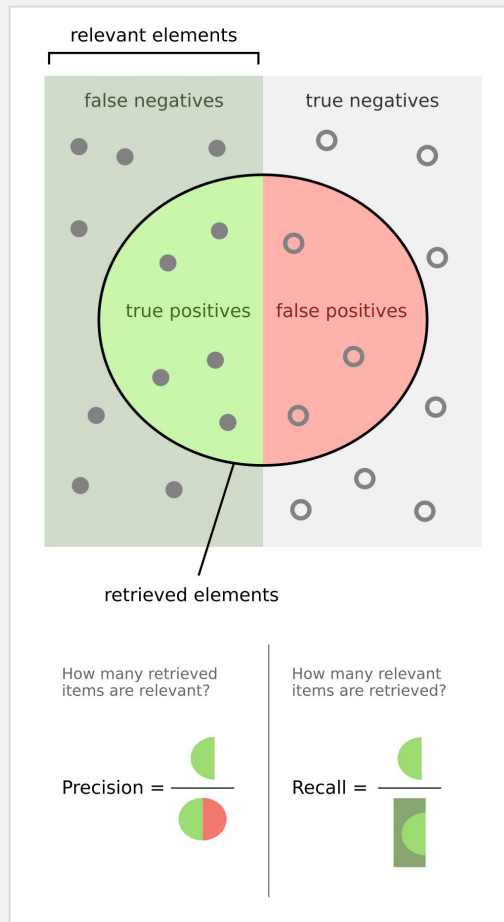
Precision & Recall

What are the precision and recall of our model?

	Actually voted	Did not vote
Pred to Vote	40	20
Pred Stay Home	15	25

$$\text{Precision} = \frac{40}{40+20} = \frac{40}{60} \approx 0.667$$

$$\text{Recall} = \frac{40}{40+15} = \frac{40}{55} \approx 0.727$$



F_1 and F_β

Another single-number measure of model success is the **F_1 score**, which is an F-measure. You might use this when accuracy is a bad choice and you want to prioritize both precision and recall equally.

$$F_1 = \frac{2}{1/\text{recall} + 1/\text{precision}} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

It is on a typical 0-to-1 scale, where 0 = bad and 1 = good

F_1 and F_β

You could also compute an **F_β score**, which is the same as an F_1 score but gives a higher (or lower) priority to recall. For example, if $\beta = 2$, you believe that recall is twice as important as precision.

$$F_\beta = (1 + \beta^2) \frac{\text{precision} \times \text{recall}}{\beta^2 \text{precision} + \text{recall}}$$

Some performance Metrics for Classification (and Information Retrieval)

What we want to know	Metric	Calculation	Examples
Of the selected items, how many are relevant?	Precision	$\frac{TP}{(TP + FP)}$	- Hiring decisions - Google searches
Of the relevant items, how many are selected?	Recall (or Sensitivity or True Positive Rate)	$\frac{TP}{(TP + FN)}$	- Medical diagnosis (cancer, covid-19) - Credit card fraud
Of all items, how many are correctly labeled as relevant / irrelevant?	Accuracy	$\frac{TP+TN}{(TP + FP + TN + FN)}$	Image classification
How accurate is the reporting (relevance and irrelevance)?	F1-score <i>(can weight relative importance of recall vs precision)</i>	$F = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$	NLP accuracy

Specificity (aka True Negative Rate) is the same calculation as Sensitivity, but for Negative values



Classification Metrics

Let's see it for ourselves!

- Open **02-classification-metrics.ipynb**



ROC AUC

Cancer Detection

- ☐ Let's return to the cancer screening problem.
- ☐ You've built a model to detect whether or not someone has cancer (positive) or not (negative). Consider the following predictions:

Cancer Detection

- Let's return to the cancer screening problem.
- You've built a model to detect whether or not someone has cancer (positive) or not (negative). Consider the following predictions:

$$\hat{p} = 0.51$$

Cancer ✓

$$\hat{p} = 0.49$$

No Cancer ✗



Why is this wrong?

Classification Thresholds

Depending on your problem, you might want to pick a **threshold** other than 0.5.

Modifying this can change (and maybe improve!) your confusion matrix.

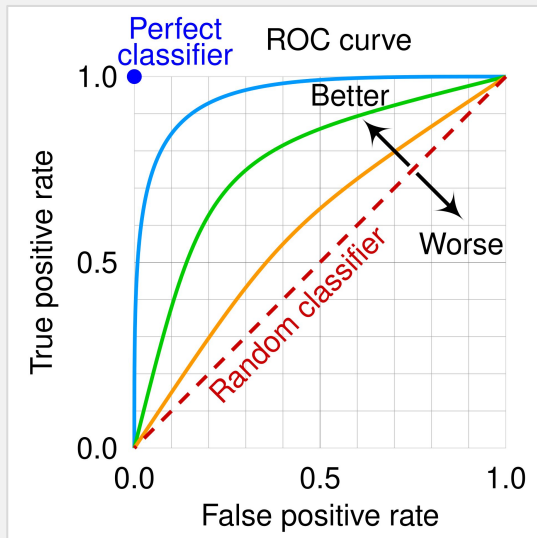
We can plot the **false positive rate (FPR)*** and **true positive rate (TPR)**** on a graph *for each possible value of our threshold!*

* aka **Specificity**

** aka **Sensitivity** or **Recall**

ROC Curve

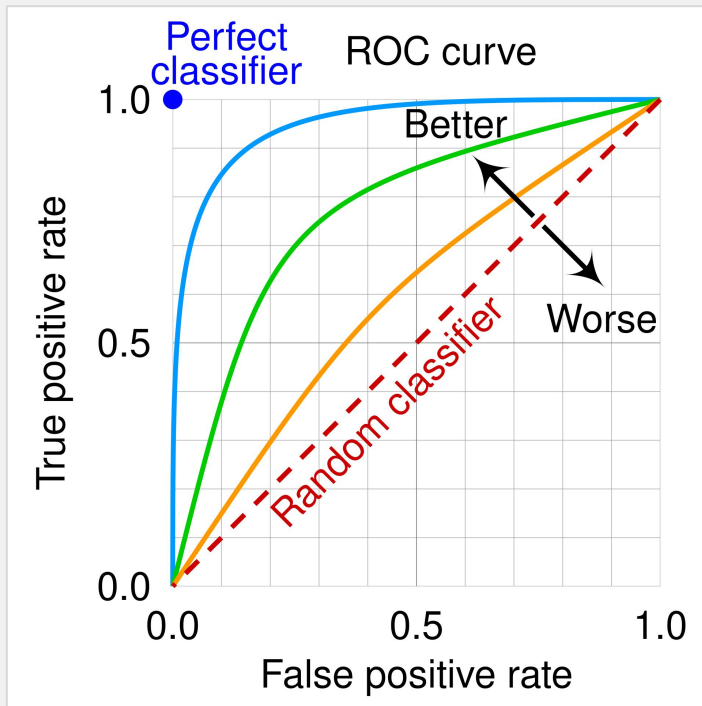
The **Receiver Operating Characteristic (ROC) Curve** is the name of this graph. It's a line plot of all the possible TPR and FPR values for every possible value of the threshold.



The AUC Score

The classifier that does total random guessing achieves a ROC curve at the 45 degree line.

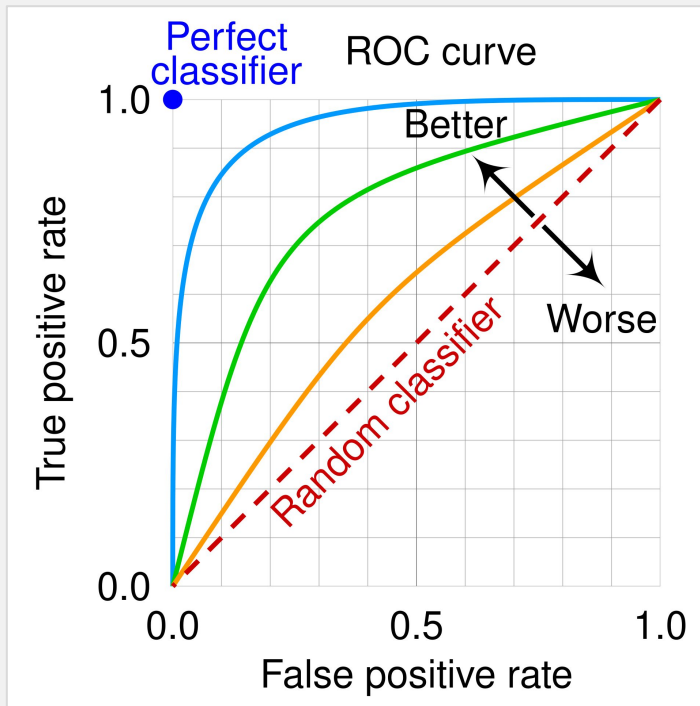
Better classifiers “bow” outward towards the perfect classifier - one with 100% TPR and 0% FPR.



The AUC Score

So, a metric we can derive from the ROC curve is to find the **area under the curve (AUC)**!

The AUC score has a maximum value of 1 and a minimum value of 0, but anything less than 0.5 would be below the “random guessing” line.



What's a "positive"?

Notice that most of the classification metrics we've learned today reference something called a **positive** and a **negative**. But consider the following example:

Again, we're doing a cancer screening, but this time, there are three options:

1. No tumor
2. Benign tumor
3. Malignant tumor

What's the "positive" outcome?

Multiclass Metrics

What's a “positive”? There isn't one! What can you use for multiclass?

- If we're doing multiclass classification, what metrics are useful to look at?

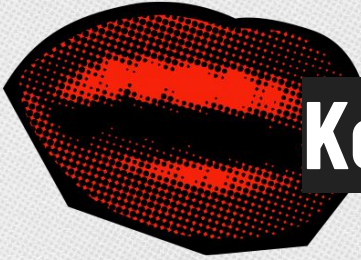
Classification Metrics

Let's see it for ourselves!

- Open **03-roc-auc.ipynb**



Bring It Home



Key takeaways



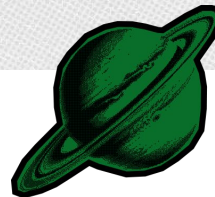
Use Classification specific metrics

Classification models use different metrics than regression models.



Get curious

Explore exactly how and when a model is making classification errors



Multiclass models

Use modified ROC AUC or other classification metrics for multiclass problems.

